CRP HEL & Wetland Screening Tool

Master Reference Guide

Comprehensive Technical & User Documentation
Version 18 (Updated June 12, 2026)

For internal use, NRCS field office coordination, and farmer outreach.
Not a substitute for official NRCS determination.

1. What Is This Tool?

The CRP HEL & Wetland Screening Tool is a pre-screening application built for farmers and NRCS conservationists to assess whether agricultural fields contain Highly Erodible Land (HEL) or wetland indicators — the two primary eligibility criteria for the Conservation Reserve Program (CRP) and USDA conservation compliance under 7 CFR Part 12.

The tool automates data collection from six official federal APIs, calculates the RUSLE2 Erosion Index (EI), classifies soils as HEL / PHEL / NOT HEL, and pre-fills official NRCS forms for submission — reducing a process that previously required manual NRCS office visits to minutes.

> ***KEY DISCLAIMER: This tool provides a pre-screening layer only. It cannot access the 1985-era frozen soils data stored in NRCS internal systems (NASIS), and therefore cannot replace an official, legally binding HEL determination made by NRCS staff. Its purpose is to help farmers and agents come better prepared before requesting a formal determination from their local NRCS field office.***

2. Farmer Mode vs. Conservationist Mode

The tool offers two distinct operating modes tailored to different user roles:

Feature	Farmer Mode	Conservationist Mode
Target User	Farmers, landowners, FSA	NRCS conservationists, agronomists
Interface	Simplified cards	Full technical detail, tabbed
HEL Result	Plain English	EI values, K/T/R/LS per component
Wetland	Traffic-light signal	Full 4-indicator table
Acreage	Total field area	Per-soil-component breakdown
Forms	AD-1026 pre-fill	NRCS-CPA-026e + CSV export
Map	Draw polygon	Polygon + satellite + soils

2A. What Farmers Can Do

- Draw their field boundary on an interactive map by clicking vertices
- Instantly see whether their field is likely HEL, Possibly HEL (PHEL), or Not HEL
- See total field acreage (SSURGO intersection method — same as NRCS GIS tools)
- View wetland risk as a simple traffic-light: Strong / Possible / Unlikely
- See a soil summary card with the dominant soil types in plain English
- Download a pre-filled AD-1026 FSA compliance certification form
- Get recommended Conservation Practice (CP) options for their land type
- Find contact info for their local NRCS field office

2B. What Conservationists Can Do

- Tab 1 — HEL Summary: RUSLE2 parameters, HEL/PHEL status, confidence, area
- Tab 2 — Wetland Analysis: 4-indicator table, hydric, vegetation, hydrology, proximity

- Tab 3 — Soil Components: K/T factors, slope, hydric rating, drainage, per-soil acres
- Tab 4 — NRCS-CPA-026e: Pre-filled official form with Sections I & II
- Tab 5 — Field Verification: Override R/K/LS/T and recalculate EI in real time

3. Technology & Architecture

Component	Technology / Description
Web Framework	Streamlit (Python)
Mapping	Leaflet.js via streamlit-folium
Soil Data	USDA SDA API — SSURGO
Elevation / Slope	USGS 3DEP 30m DEM via py3dep
R-Factor (Primary)	NOAA CONUS 800m raster via rasterio
R-Factor (Fallback 1)	NOAA CDO GHCND stations
R-Factor (Fallback 2)	State-level NRCS FOTG (50 states)
Vegetation	USGS NLCD 2021 raster
Wetland Proximity	FWS NWI WMS
Acreage	Shapely + pyproj EPSG:5070
Geocoding	Nominatim (OpenStreetMap)
PDF Generation	ReportLab
Concurrency	Python ThreadPoolExecutor
Deployment	Render (cloud) / Local

4. RUSLE2 Erosion Index Formula & HEL Determination

4A. Core Formula

$$EI = (R \times K \times LS) / T$$

R = Rainfall Erosivity (MJ·mm/ha·h/year) | **K** = Soil Erodibility (t/MJ·mm) | **LS** = Slope factor (dimensionless) | **T** = Soil Loss Tolerance (t/acre/year)

4B. HEL Classification Thresholds

Classification	Condition	Action
HEL	EI_{min} ≥ 8.0	Field is HEL — CRP eligible
PHEL	EI_{min} < 8.0 AND EI_{max} ≥ 8.0	Requires field visit
NOT HEL	EI_{max} < 8.0	Ineligible

4C. PHEL Logic & Example

If slope range 6–10%: EI at 6% = 7.69 (below 8.0) and EI at 10% = 16.23 (above 8.0) → **PHEL (requires NRCS verification)**

5. Official Data Sources & APIs

Data	Source	API/Method	Accuracy
K-Factor	SSURGO kwfact	SDA API, 0cm, majcomp	Official NRCS
T-Factor	SSURGO tfact	SDA API, majcomp	Official NRCS
R (1st)	NOAA CONUS 800m	Rasterio pixel lookup	±1–3%
R (2nd)	NOAA CDO GHCND	Brown & Foster formula	±5–8%
R (3rd)	NRCS FOTG Handbook	State table, 50 states	±20–30%
DEM/Slope	USGS 3DEP 30m	py3dep library	±5% LS error
NLCD/Vegetation	USGS NLCD 2021	Raster pixel lookup	30m resolution
Wetland Proximity	FWS NWI	WMS tiles	Polygon distance
Acreage	SSURGO mupolygon	SDA STAsText() intersection	909.4 ac validated
County/State	Nominatim OSM	Reverse geocoding	City-level accuracy

6. Fallback Chains & Resilience

R-Factor (4 Tiers): ① NOAA CONUS 800m raster (±1–3%) → ② NOAA CDO GHCND (±5–8%) → ③ State FOTG table (±20–30%) → ④ National default (±20–30%)

LS-Factor: ① USGS 3DEP DEM-based (±5–15%) → ② Slope$^{1.2}$ × 0.1 approximation (±23%)

Acreage: ① SSURGO mupolygon intersection → ② Polygon ÷ component proportions

Performance: All three primary APIs run concurrently (ThreadPoolExecutor), reducing first load from 45s to 20s (55% faster).

7. NRCS Compliance Status

Component	Status	Notes
EI formula: R×K×LS/T	FULLY COMPLIANT	Exact: 7 CFR § 12.21(a)(2)
HEL threshold ≥8.0	FULLY COMPLIANT	NRCS Part 616
K-Factor (SSURGO)	FULLY COMPLIANT	Surface, majcomp only
T-Factor (SSURGO)	FULLY COMPLIANT	Majcomp, no transform
R-Factor (raster)	COMPLIANT	±1–3% vs Ag Handbook 703
S-Factor	FULLY COMPLIANT	Both regimes match RUSLE2
L-Factor	PARTIAL	Simplified; ±15–40%
Acreage method	FULLY COMPLIANT	= NRCS GIS method
CPA-026e PDF	COMPLIANT	Matches NRCS form (8/2013)
Overall	~85%	Suitable pre-screening; not official

Regulatory References

- 7 CFR § 12.21 — Identification of Highly Erodible Lands
- 7 CFR Part 12 — Conservation Compliance
- NRCS Technical Soil Services Handbook, Part 616
- NRCS Agriculture Handbook 703 — R-Factor data

8. Official USDA Forms Auto-Generated

8A. NRCS-CPA-026e — HEL & Wetland Determination

Purpose: Official NRCS form documenting HEL determination with RUSLE2 data.
Auto-fills: County, State, Request Date, Section I (HEL results), Section II (Wetlands), RUSLE2 parameters.
Requires: *NRCS staff signature and field verification before official submission.*

8B. AD-1026 — HELC and Wetland Certification

Purpose: FSA compliance certification form filed by farmer.
Auto-fills: Producer info, HEL status, Wetland status, Date.
Requires: *Producer signature; filed with local FSA office.*

9. Remaining Gaps (as of June 12, 2026)

Critical Gaps:

1. NOAA CDO single-year data: Uses 2023 only; should be 22+ years per Wischmeier & Smith 1978. Effort: Medium.

2. LS flow simplification: Not proper D8 algorithm; ±15–40% error on complex terrain. Effort: High.

3. Sodbust Y/N blank: Cannot auto-determine from public data. By design; requires NRCS lookup.

Moderate/Minor Gaps:

4. NRCS Office locator is stub (Effort: Low)

5. CP suggestions static; static recommendations (Effort: High)

6. No unit tests (Effort: Medium)

7. No audit trail (Effort: High)

8. Mobile layout not optimized (Effort: Medium)

Note on C-Factor & P-Factor

Not a gap. C-Factor and P-Factor are **NOT required for HEL determination** per 7 CFR 610.14. They are used in conservation *practice planning* (selecting which CP practices reduce erosion), not for HEL *determination* (screening which fields qualify). This tool focuses on HEL screening only, using the formula: **EI = (R × K × LS) / T** (R-factor, K-factor, LS-factor, T-factor only). See CORRECTION_NOTICE_C_P_FACTORS.md for full explanation.

10. Feature Completion Status

Feature Area	%
HEL/RUSLE2 Core (R, K, T)	95%
Raster R-Factor (NOAA 800m)	95%
LS Factor (DEM-based)	75%
SSURGO Integration	95%
NOAA CDO fallback	70%
Wetland Indicators	85%
Two-Tier UI	95%
CPA-026e PDF	95%
AD-1026 FSA	85%
CP Practice Suggestions	40%
Map & Location	95%
Confidence Flags	90%
Area Calculation	95%
Performance Optimization	90%
NRCS Office Locator	5%
Test Coverage	0%
OVERALL	**~83%**

11. Field Validation Against Real SSURGO Data

Region	Test	EI/Result	Status
Iowa (Boone)	Steep soils	EI 97.97	HEL — Correct
Nebraska	Moderate slopes	EI 50.41	HEL — Correct
New York	Mixed 2–50%	EI 101.95	PHEL/HEL — Correct
Texas	Very steep	EI 138.08	HEL — Correct
Colorado	Rolling	EI 9.87	PHEL — Correct
Iowa polygon	Field boundary	909.4 acres	SSURGO intersection validated
Iowa R-factor	Boone coords	R = 155.2	Expected ~160, within ±3%

This document was generated from verified project files.
All Compliant items tested against real SSURGO data.
Updated June 12, 2026 — C-Factor & P-Factor clarified as non-gap per 7 CFR 610.14